RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

Final Project Report

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Internship: Data Analyst Internship

Company: Data X Labs

Tools Used: Python, SQL, Power BI, Pandas, Matplotlib, Seaborn, Jupyter Notebook

Dataset: Walmart Sales Dataset

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1. Introduction

Retail businesses generate large volumes of sales data every day. Analyzing this data helps organizations understand customer behavior, identify profitable products, monitor sales performance, and make informed business decisions.

The objective of this project is to analyze Walmart sales data using SQL, Python, and Power BI to evaluate business performance, identify sales trends, and generate meaningful insights through data visualization and interactive dashboards.

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2. Objectives

The primary objectives of this project are:

- Analyze retail sales performance using SQL and Python.
- Identify monthly sales trends and customer purchasing behavior.
- Compare sales performance across different cities and product lines.
- Develop an interactive Power BI dashboard for business reporting.
- Generate actionable business insights to support decision-making.

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3. Dataset Description

The Walmart Sales Dataset contains transactional information collected from retail stores. The dataset includes details such as invoice ID, branch, city, customer type, gender, product line, quantity purchased, total sales amount, gross income, customer ratings, date, and time of purchase.

Before analysis, the dataset was cleaned and prepared to ensure accurate reporting and visualization.

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4. Tools and Technologies

- Python (Pandas, Matplotlib, Seaborn)
- SQL (MySQL)
- Power BI
- GitHub
- Google Colab

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5. Analysis Performed

The following analyses were carried out during this project:

- Data exploration and statistical summary
- Missing value analysis
- Sales analysis by product line
- Monthly sales trend analysis
- City-wise sales comparison
- Customer type profitability analysis
- SQL-based business queries
- Interactive dashboard development using Power BI

The dashboard includes KPI cards, monthly sales trends, product-wise performance, city-wise analysis, and interactive filters for business users.

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6. Key Findings

The analysis revealed several important business insights:

- Naypyitaw generated the highest overall sales revenue.
- Evening sales contributed more revenue than morning and afternoon sales.
- Food and Beverages and Fashion Accessories were among the top-performing product categories.

- Member customers generated higher gross income compared to normal customers.
- Sales performance varied across cities, months, and product lines.

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7. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Increase inventory for top-performing product categories.
- Focus promotional campaigns in high-performing cities.
- Encourage customer membership programs to improve profitability.
- Use monthly sales trends for better inventory planning and resource allocation.
- Monitor low-performing product categories and implement targeted marketing strategies.

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8. Conclusion

This project successfully demonstrated how SQL, Python, and Power BI can be integrated to analyze retail business performance. Through data exploration, visualization, and dashboard development, meaningful business insights were generated to support data-driven decision-making.

The project also strengthened practical skills in SQL querying, exploratory data analysis, business intelligence, dashboard development, and data visualization, providing valuable experience in solving real-world retail business problems.